“Mental Health Analysis Using Machine Learning Models”

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(Specialization in Artificial Intelligence & Machine Learning / Data Science)

Submitted by

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Visakhapatnam
2026

DECLARATION

I hereby declare that the project report entitled “Mental Health Analysis Using Machine Learning Models” is an original work done in the Department of Artificial Intelligence and Data Science, GITAM School of CSE, GITAM (Deemed to be University), submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering (AI&ML/DS). The work has not been submitted to any other college or University for the award of any degree or diploma.

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CERTIFICATE

This is to certify that the project report entitled “Mental Health Analysis Using Machine Learning Models” is a bonafide record of work carried out by M. V. S. Satya Mrudula (2023000597), G. Rishikar Reddy (2023001672), D. Vivek (2023002218) and T. S. L. Priyanka (2023004539) students submitted in partial fulfillment of requirement for the award of degree of Bachelor of Technology in Computer Science and Engineering (AI&ML/DS).

Date :

Project Guide Head of the Department

ACKNOWLEDGEMENT

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We are grateful to the Head of the Department and to all faculty members of the Department of Artificial Intelligence and Data Science for the environment and resources that made this project possible. We also thank the project coordinators for arranging the review process.

Finally we are thankful to families and friends for families and friends patience and support, throughout this project.

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1. ABSTRACT

Mental health concerns often go unnoticed until people speak up about symptoms or get a check. Early screening can spot those who might be at risk for depression, anxiety and stress.. Today screening mostly uses questionnaires and the eye of a clinician. This project looks at whether machine-learning models can mix data about a person's age, habits, lifestyle, school or work situation, friends and feelings to help screening. The plan is to use two data streams. One stream will take the questionnaire answers and feed them into an FT-Transformer. The other stream will take the free-form text and feed it into DeBERTa. The two model outputs will then be joined to give predictions, for depression, anxiety and stress. The goal is to explain what drives those predictions so that screening and professional decisionsre clearer. At this point both data sets have been. The models chosen. Next steps are to clean the data build the models merge them and test how well they work.

2. INTRODUCTION

2.1 Background

Mental health is an important part of being healthy overall but conditions like depression, anxiety and stress are often not noticed. A lot of people do not ask for help until their symptoms get bad and regular checking for these issues is not always possible in schools, colleges or offices. Identifying people who might be at risk on can help them get the right care quickly.

Machine learning can help with this kind of checking by looking for patterns in data that show a persons situation and experiences. Questionnaires collect information about age, habits, how someone lives, their school or work life and their social life in a way.. Written words from a person, like how they describe their feelings or thoughts can tell us things that a fixed questionnaire might miss. Looking at both of these together can give an understanding than looking at either one alone.

2.2 Need for the System

Screening that relies on a questionnaire can miss important information and clinical records are not always available, outside health services. Faculty, counsellors and workplace support teams need a way to make the screening process more organized. A system that uses both responses and written psychological text and that produces clear easy to read outputs can help these groups make better decisions while letting trained professionals make the final judgment.

2.3 Justification of the Title

The title "Mental Health Analysis Using Machine Learning Models" explains the parts of the work. The machine-learning models learn how the input data relates to the screening outcomes of depending on fixed scoring rules. The analysis focuses on health, especially the three conditions of depression, anxiety and stress. The word "analysis" shows that the system looks at the inputs to estimate risk and that the outputs are meant to help with decisions not to act as a diagnosis.

2.4 Scope of the Project

The project covers the selection and preparation of the questionnaire and psychological text datasets, the choice of FT-Transformer and DeBERTa as the two analysis models, the combination of their outputs, and the evaluation of the resulting predictions. It does not cover clinical diagnosis, treatment recommendations or the deployment of the system in a live clinical setting. Predictions are intended only to support screening and professional decision-making.

3. LITERATURE REVIEW

3.1 Machine Learning for Depression and Anxiety Prediction

Several studies have explored the use of machine learning to identify symptoms related to depression, anxiety and stress. ShamsEldin et al. (2025) [1] investigated the prediction of these three conditions using psychometric data. Their study used 39,775 anonymised responses collected through the DASS-42 questionnaire along with demographic information. Five classification algorithms, including Support Vector Machine (SVM), Random Forest, Decision Tree, K-Nearest Neighbours and Naive Bayes, were compared. SVM achieved the highest reported accuracy, with 99.3% for depression, 98.9% for anxiety and 98.8% for stress. The study demonstrates how questionnaire responses can be used for mental health screening. However, the reliability of questionnaire-based labels and the need for external validation must also be considered.

Nemesure et al. (2021) [2] Examined how well depression and anxiety could be predicted with electronic health records. Their method mixed machine learning models and applied SHAP (SHapley Additive exPlanations) to see which features mattered most for the predictions. The reported ROC-AUC numbers were 0.67 for depressive disorder (MDD) and 0.73 for generalised anxiety disorder (GAD). These results show that data, about demographics and biomedical factors can give clues to spot mental health risks. However the moderate prediction performance means

that more work and further validation are needed to improve the prediction of depression and anxiety.

3.2 Behavioural and Long-Term Mental Health Prediction

Strunk (2023) [3] examined how anxiety and depression symptoms could be predicted during the COVID-19 pandemic. I am pleased that Strunk followed 1,200 adults from the United States and studied 68 variables. Using Elastic Net regression with repeated cross-validation, Bailey and Strunk identified the predictors. Their results showed that the test R^2 value for anxiety was 0.7818 and the test R^2 value for depression was 0.7691, which's a strong match. The study emphasized that stress, sleep and psychological vulnerabilities matter a lot when we analyze health. However I notice that the specific setting and the limits of follow-up may make the findings less applicable to populations.

Ahmed (2023) [4] proposed a system for depression screening using smartphone application usage data. Their approach, called Mon Majhi collected seven days of application usage history and used machine learning to identify possible depression-related patterns. The study involved 100 students and used LightGBM along with feature selection techniques such as Boruta and SHAP for model interpretation. The reported recall was 82.4% precision was 75.0%. F1-score was 78.5%. This research shows that everyday behavioural patterns can provide information, for mental health screening. However the small sample size limits the extent to which the results can be generalised.

3.3 Transformer Models for Questionnaire and Psychological Text Analysis

Advances in learning have created transformer-based models that can work with many kinds of data. Two of these models are DeBERTa. Ft-transformer and they are the basis for the approach we propose in this project.

He and colleagues in 2020 presented DeBERTa, which stands for BERT with Disentangled Attention. Unlike BERT models DeBERTa handles the meaning of words and their positions separately using disentangled attention. This allows DeBERTa to grasp how words relate to each other inside a sentence. Because psychological text often includes feelings, ideas and personal stories we chose DeBERTa to analyse the text in this project.A 2021 study titled Revisiting Deep Learning Models for Tabular Data looked at deep learning methods for data. Introduced the

FT-Transformer, which means Feature Tokenizer Transformer. FT-Transformer turns each attribute into a token. FT-Transformer feeds these tokens through transformer layers. This makes FT-Transformer a good choice, for examining questionnaire answers that include demographics, behaviours and psychological factors. In our work we use FT-Transformer to handle questionnaire data and spot patterns linked to depression, anxiety and stress.

Both models serve input types. DeBERTa is dedicated to understanding text while FT-Transformer deals with questionnaire data. DeBERTa and FT-Transformer's distinct functions fit well with the two input streams we plan to use in this project.

3.4 Summary and Research Gap

Table 3.1 summarises the reviewed work and the gap that this project addresses.

Table 3.1: Summary of Reviewed Literature

Ref.	Research Focus	Approach / Model	Main Contribution	Limitation
[1]	Psychometric screening	SVM, Random Forest, Decision Tree, KNN, Naive Bayes	Prediction of depression, anxiety and stress using DASS-42	Self-reported labels and need for external validation
[2]	Health-record-based prediction	Stacked ML, XGBoost, SHAP	Prediction of depression and anxiety using routine health records	Moderate discrimination and limited generalisation
[3]	Long-term symptom prediction	Elastic Net	Identification of baseline factors associated with symptoms over 12 months	Pandemic-specific context and follow-up limitations
[4]	Smartphone-based screening	LightGBM, Boruta, SHAP	Depression screening using smartphone application activity	Small sample size and questionnaire-based labels
[5]	Psychological text processing	DeBERTa	Disentangled attention for improved contextual text representation	Applied to psychological text analysis in this project
[6]	Structured data processing	FT-Transformer	Transformer-based architecture for tabular data	Applied to questionnaire analysis in this project

The literature shows that prediction of these conditions is established, but the two kinds of input are usually studied separately. This project therefore combines structured questionnaire responses and psychological text in one framework and predicts all three conditions.

4. PROBLEM IDENTIFICATION & OBJECTIVES

4.1 Problem Identification

Mental-health concerns may remain unrecognised until individuals report symptoms or receive professional evaluation. The problem addressed in this project is whether machine-learning models can combine demographic, behavioural, lifestyle, academic or workplace, social and psychological factors to identify individuals who may be at risk of depression, anxiety or stress, and whether they can do so in a way that explains the contributing factors.

4.2 Aim of the Project

The aim is to investigate whether machine-learning models can support early screening for possible depression, anxiety and stress by using relevant demographic, behavioural, lifestyle, academic or workplace, social and psychological factors.

4.3 Objectives

- Analyse questionnaire data using the FT-Transformer model.
- Analyse psychological text data using the DeBERTa model.
- Combine the outputs of both models to generate predictions for depression, anxiety and stress.

4.4 Feedback from Review 1 and Actions Taken

The feedback received from the guide and reviewers in Review 1 [7] was used to shape the current scope. Table 4.1 lists the feedback and the corresponding actions.

Table 4.1 : Review 1 feedback and actions taken

S.No.	Feedback received in Review 1	Action taken
1	Expand the project beyond questionnaire-based analysis.	Added psychological text data as an additional source alongside questionnaire data.
2	Avoid relying on older models such as SVM and Logistic Regression; explore updated models from research papers.	Selected DeBERTa for text analysis and FT-Transformer for questionnaire analysis.
3	Include only information that the team can clearly explain and justify.	Planned two separate models whose outputs are combined to predict depression, anxiety and stress.

4.5 Requirement Analysis

The requirements derived from the objectives and from the Review 1 feedback are listed in Table 4.2.

Table 4.2 : Functional requirements of the system

S.No.	Requirement	Purpose
1	Use the benchmark questionnaire and psychological text datasets.	Provide the inputs for both models
2	Inspect data formats, missing values, labels and data quality.	Ensure the data are fit for modelling
3	Preprocess structured questionnaire responses and psychological text.	Prepare each stream for its model
4	Analyse questionnaire data using FT-Transformer.	Produce the questionnaire-based output
5	Analyse psychological text using DeBERTa.	Produce the text-based output
6	Combine model outputs to predict depression, anxiety and stress.	Give one result per condition
7	Provide clear and readable outputs.	Make results usable by screening staff

5. EXISTING SYSTEM

5.1 Current Approaches

Screening for depression anxiety and stress is commonly carried out with questionnaires that are scored manually or with simple statistical rules. Where machine learning has been applied many studies use a data source, typically questionnaire or clinical records together with conventional

classifiers such, as Support Vector Machines and Logistic Regression. Table 5.1 summarises these approaches and their drawbacks.

Table 5.1 : Existing approaches and their drawbacks

Existing approach	Main drawback
Standardised questionnaire scoring	Relies on one structured source; written expression of feelings is not used.
Questionnaire-based machine learning with SVM or Logistic Regression	Older models that may not capture complex relationships as effectively as recent architectures.
Clinical-record prediction	Depends on records that are not available for many screening settings.
Single-source prediction	Predictions are not combined with a second, complementary source of information.

Existing methods often depend on one data source and an older model. They usually don't mix information with psychological text in a way that is easy to read. Most of the time the results are not combined into a clear output.

6. PROPOSED SYSTEM

6.1 Overview

The proposed system extends questionnaire-based analysis by adding text as a second data source. Two separate models are used, each suited to its data type. An FT-Transformer analyses the questionnaire responses and DeBERTa analyses the psychological text. The outputs of the two models are then combined to generate predictions for depression, anxiety and stress.

6.2 Modules of the Proposed System

- Data validation: checks the questionnaire and text datasets for data formats missing values, labels and data quality.
- Questionnaire pipeline: preprocesses the responses and analyses them with the FT-Transformer.
- Text pipeline: preprocesses the psychological text and analyses it with DeBERTa.
- Integration module: combines the outputs of both models to produce predictions for depression, anxiety and stress.

- Output module: presents the predictions, in an readable form.

6.3 Datasets

Two datasets are used in this project as shown in Table 6.1.

The dataset collection is finished.

The next step is to confirm the labels check the data formats and resolve any data-quality issues.

Table 6.1 : Datasets used in the project

Dataset	Content	Use in the project
Benchmark questionnaire dataset	Responses covering demographic, behavioural, lifestyle, academic or workplace, social and psychological factors	Input to the FT-Transformer for questionnaire analysis
Psychological text dataset	Written psychological text	Input to DeBERTa for text analysis

6.4 Proposed Methodology

The questionnaire data and the psychological text data each travel through their separate pipelines, a process that helps keep the information clear.

The FT-Transformer examines the questionnaire data while DeBERTa examines the text and the outputs from the FT-Transformer and DeBERTa are then merged to produce predictions, for depression, anxiety and stress.

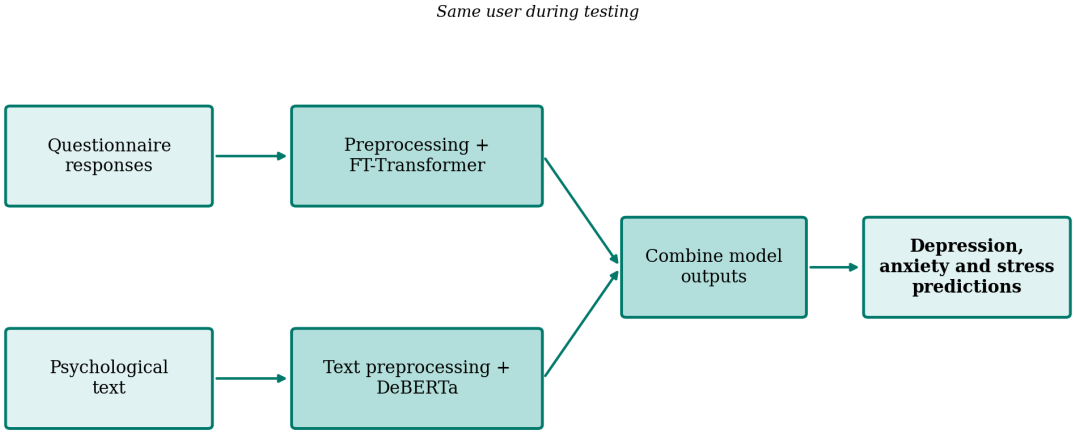


Figure 6.1 : Proposed methodology

6.5 Combination of Model Outputs

During testing the questionnaire responses and psychological text come from the user. Each model gives its output. Then in the integration stage these outputs are put together to form one prediction for each condition. The exact way we combine them will be decided after running baseline experiments. We'll share that method along, with the results.

6.6 Evaluation Plan

We will compare the models using metrics that work well for screening tasks (see Table 6.2). The evaluation will also look for missing data, uneven class distribution, data leakage and whether error rates differ across subgroups.

Table 6.2 : Planned evaluation of the models

Task	Model	Planned metrics
Questionnaire-based prediction of depression, anxiety and stress	FT-Transformer	Accuracy, precision, recall, F1-score, ROC-AUC
Text-based prediction of depression, anxiety and stress	DeBERTa	Accuracy, precision, recall, F1-score, ROC-AUC
Combined prediction	Integration of both model outputs	Same metrics, compared with each single-model output

6.7 Improvement over the Existing System

Table 6.3 compares the existing and the proposed system.

Table 6.3 : Comparison of the existing and the proposed system

Aspect	Existing system	Proposed system
Data used	Single structured source, or clinical records	Questionnaire responses and psychological text
Models	Support Vector Machine, Logistic Regression	FT-Transformer and DeBERTa
Predictions	Usually one condition at a time	Depression, anxiety and stress
Output	Score or category with little context	Clear and readable outputs for screening staff
Decision making	Depends on the individual assessor	Decision support, with professional review of results

7. SYSTEM ARCHITECTURE

7.1 Project Workflow

The implementation follows the workflow shown in Figure 7.1.

The questionnaire and psychological text pipelines run separately. Their outputs meet in the combined output stage.

Table 7.1 shows the status of each stage at the time of this review.

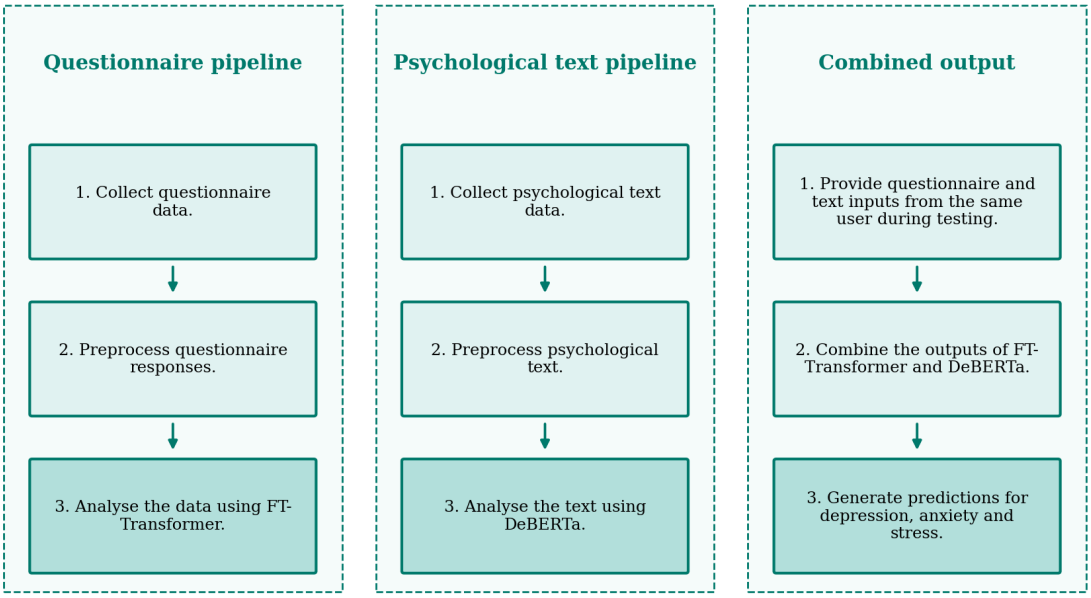


Figure 7.1 : Proposed project workflow

- Validation checkpoints: verify that the data formats are correct that no values are missing that labels are accurate and that overall data quality is good before each analysis stage.
- Expected deliverables: a questionnaire, text datasets, trained FT-Transformer and DeBERTa models combined predictions, for depression, anxiety and stress and results that can be reproduced for the next review.

Table 7.1 : Project workflow and status

Stage	Activity	Status
1	Collection of the questionnaire and psychological text datasets	Completed
2	Data inspection and preprocessing of both datasets	Next step
3	Model development with FT-Transformer and DeBERTa	Planned
4	Integration of model outputs	Planned
5	Testing and evaluation	Planned
6	Presentation of results and review	Planned

7.2 Data Flow Diagram

Figure 7.2 shows the way data moves from one part to another. While testing the same user fills out a questionnaire. Also provides some psychological text. The answers from the questionnaire are cleaned up. Given to the FT-Transformer. The psychological text is also cleaned up. Given to DeBERTa. The results, from both models are then put together to predict depression, anxiety and stress. Table 7.2 lists the parts that make up the data flow.

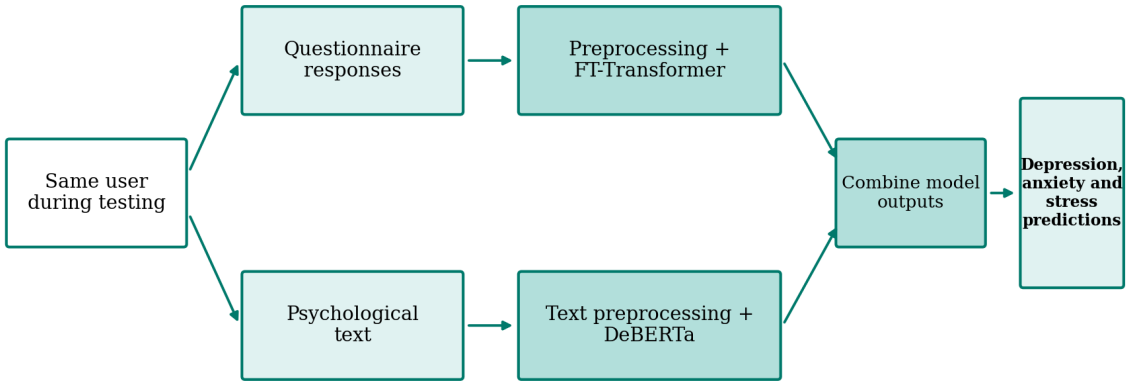


Figure 7.2 : Conceptual data flow diagram of the proposed system

Table 7.2 : Main components of the data flow

Component	Description
Input	Questionnaire responses and psychological text from the same user
Processing	Preprocessing of each data stream and data-quality checks
Prediction	Analysis with the FT-Transformer and DeBERTa
Integration	Combination of the two model outputs
Output	Predictions for depression, anxiety and stress

7.3 Use Case Diagram

Figure 7.3 shows the conceptual use case design. I notice that the user is the actor. Requesting a mental-health screening involves analysing the questionnaire responses, with the FT-Transformer analysing the text with DeBERTa and combining both predictions. Table 7.3 lists the use cases.

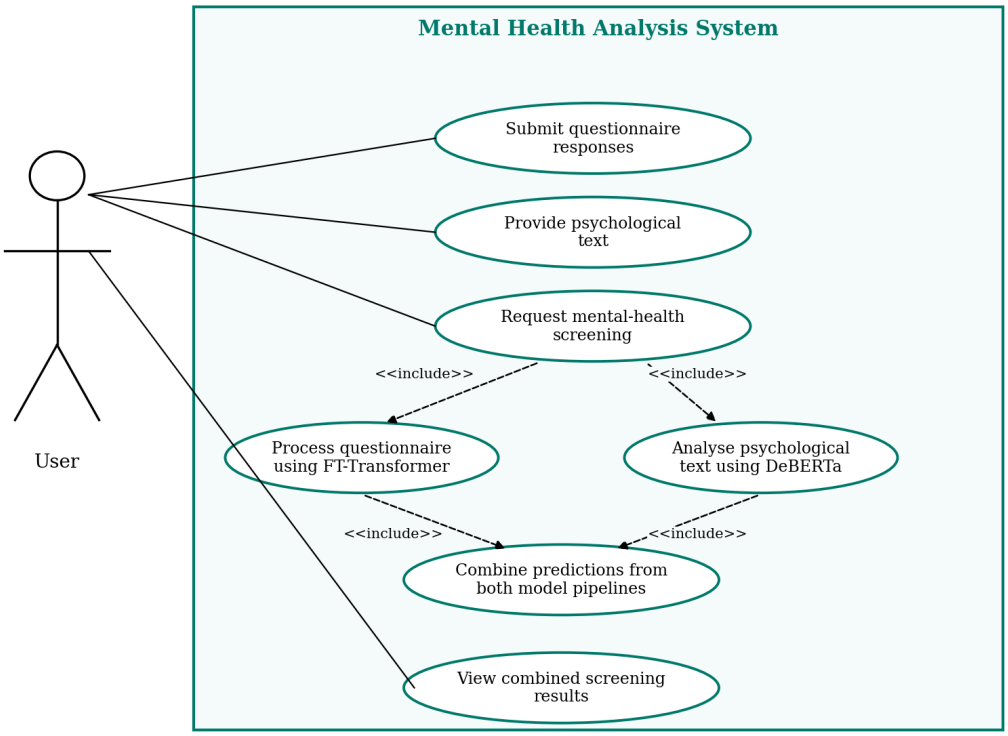


Figure 7.3 : Use case diagram of the proposed system

Table 7.3 : Use cases of the system

Use case	Description
Submit questionnaire responses	The user provides the structured questionnaire responses
Provide psychological text	The user provides written psychological text
Request mental-health screening	The user starts the screening, which includes the two analyses below
Process questionnaire using FT-Transformer	Included in the screening request
Analyse psychological text using DeBERTa	Included in the screening request
Combine predictions from both model pipelines	Included in the screening request
View combined screening results	The user sees predictions for depression, anxiety and stress

7.4 Sequence Diagram

Figure 7.4 shows how the user, the frontend interface the two analysis models and the integration module work together. The user fills out the questionnaire adds the text and then asks for a

screening. The frontend interface takes each of these inputs. Sends them to their respective models. Each model makes a prediction based on its input. Then both predictions go to the integration module. The integration module puts them together to create a result. That combined result is sent back to the user. It's a step-, by-step process. User gives input models analyse, module combines user gets results.

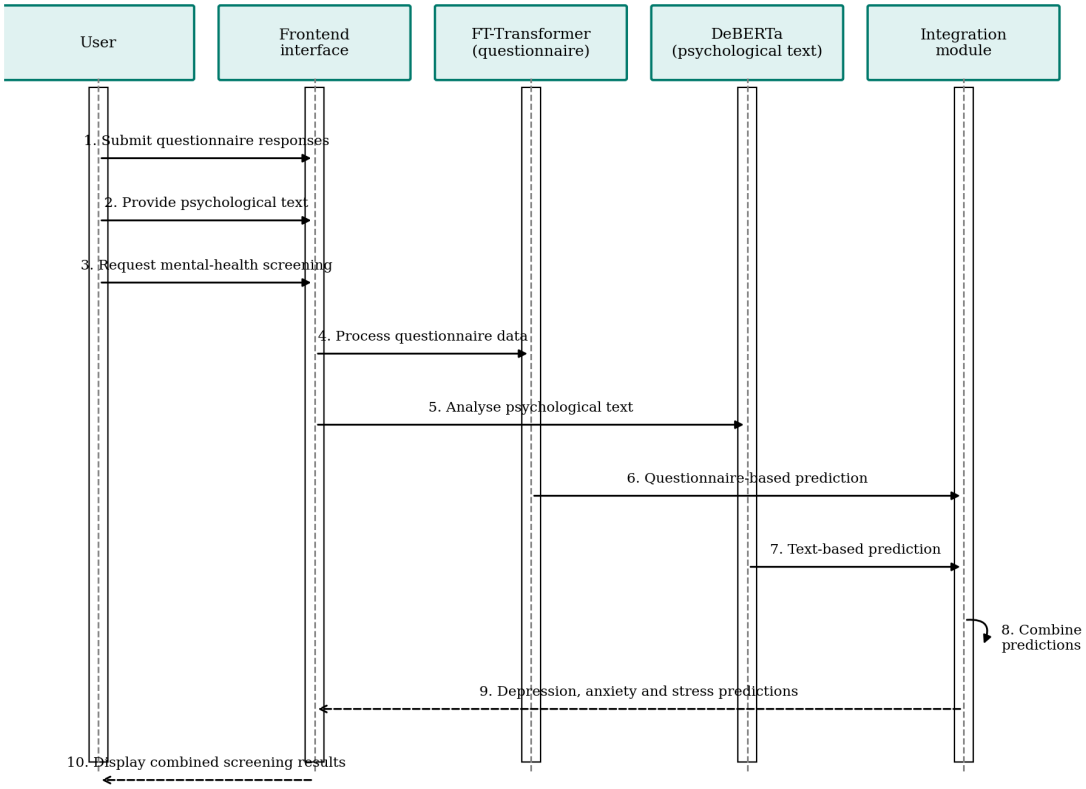


Figure 7.4 : Sequence diagram of the mental-health screening process

7.5 Activity Diagram

Figure 7.5 shows the activity of the screening process across four swimlanes. After the user submits both inputs and requests a screening the questionnaire responses and the psychological text are preprocessed in parallel. The questionnaire data are checked for validity; if the check fails the record is flagged for correction. Otherwise the FT-Transformer analyses the data. DeBERTa analyses the text. The integration stage then combines the two outputs generates predictions, for depression, anxiety and stress and displays them to the user.

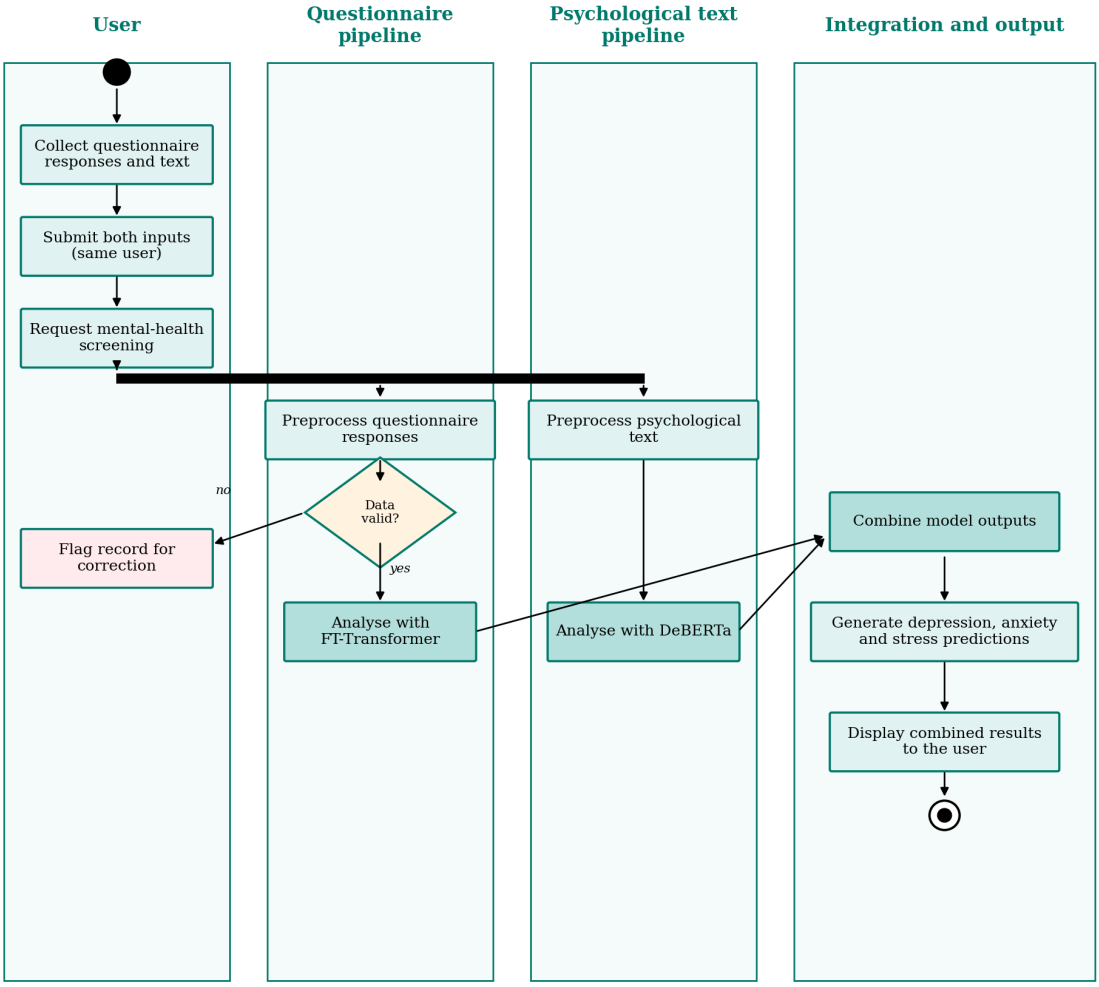


Figure 7.5 : Activity diagram of the mental-health screening process

8. TOOLS/TECHNOLOGIES USED

8.1 Tools

The tools selected for the project are listed in Table 8.1 along with their significance in the work.

Table 8.1 : Tools used and their significance

Tool	Significance in the project
Python	Main programming language for data processing and model development
Pandas	Loading, cleaning and transforming the datasets
NumPy	Numerical operations on arrays and feature matrices
Matplotlib	Plotting of data distributions and results
Seaborn	Statistical visualisation for exploring data quality and relationships
Scikit-learn	Data splitting, evaluation metrics and supporting utilities

8.2 Algorithms

The two analysis models selected for the project are listed in Table 8.2.

Table 8.2 : Algorithms used and their role

Algorithm	Input data	Role in the project
FT-Transformer	Questionnaire data	Analyses structured responses for screening
DeBERTa	Psychological text	Analyses written psychological text for screening

9. CONCLUSION

9.1 Conclusion

This project looks at whether machine-learning models can help with screening for depression, anxiety and stress by mixing questionnaire data with psychological text. The FT-Transformer is used to analyse questionnaire data while DeBERTa is used to analyse text. The outputs from the FT-Transformer and DeBERTa are combined to create predictions for depression, anxiety and stress. These predictions are meant to give decision-support information for screening and for review.

The work completed far includes the following:

- The collection of the questionnaire dataset and the psychological text dataset is finished.
- The choice of models is finished: FT-Transformer for questionnaire data and DeBERTa, for text.

- The planned methodology and workflow are ready.

9.2 Work Plan and Timeline

The work plan is given in Table 9.1. The timeline is provisional and will be updated to match the actual progress and the schedule approved by the guide.

Table 9.1 : Work plan and timeline

Task	Task owner	Status
Problem identification and scope	Team	Done / reviewed
Literature survey and research gaps	Team	Done / reviewed
Project objectives and workplan	Team	Done / reviewed
Requirement analysis and dataset audit	Assigned members	Done / reviewed
Tools, algorithms and system design	Assigned members	Planned
Proposed methodology and workflow	Team	Planned
Baseline experiments and evaluation	Assigned members	Planned
Phase 1 closure / project exhibition	Team	Planned

9.3 Next Actions

The next actions are to confirm the dataset and permissions, finalise the labels and feature dictionary, implement the baseline models, and present reproducible results at the next review. Data preprocessing is the next step, followed by model development, integration, testing and evaluation.

9.4 Disclosure Statements

- I will maintain academic honesty by doing my own work and avoiding plagiarism, cheating, unauthorised collaboration and the falsification of information.
- I will properly acknowledge and cite all sources I use, including information from the internet, books, software and AI tools.
- I will use institutional resources responsibly and only for authorised academic or research activities. I understand that I am responsible for my own actions and decisions.

10. REFERENCES

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